

Strong Pillar 1 at four tail positions of $(2, 2, 1, 1, 1)$: operator vanishing via a row-1-pair embedding

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28 May 2026 (third prove session)

Abstract

Pursuing the strong per-subset operator vanishing conjecture for $\lambda = (2, 2, 1^m)$, we prove that the factor-removed tail chains $\Omega_{\text{tail}}^{(\lambda)}[\widehat{p}]$ act as the *zero operator* on V^λ for the four positions $p \in \{12, 13, 16, 17\}$ at $\lambda = (2, 2, 1, 1, 1)$. This complements the two structurally proved positions ($p = 10, 15$) from [2], leaving five open positions ($p \in \{11, 14, 18, 19, 20\}$) where Strong Pillar 1 ($\subseteq E_1^-$) is verified computationally but the structural proof remains incomplete. The new identities are:

- (K1) $S_1 S_2 = q \cdot I$ on $E_1^+|_{V^\lambda}$, by direct Hoefsmit computation using the unique axial distance $d = -2$.
- (K2) $S_1 R'_n = q \cdot S_{n-1} S_{n-2} \cdots S_3 \cdot S_1$ as operators on V^λ , for any $n \geq 4$, by commuting S_1 to the right and applying (K1).
- (K3) Lemma A: $S_1 R'_7 \cdot V^\lambda \subseteq E_3^- \cap E_4^-|_{V^\lambda}$, by Pillar 1 applied to the embedded sub-representation $E_1^+|_{V^\lambda} \cong V^{(2,1,1,1)}$.
- (K4) Lemma B: $(R'_5)^2 \cdot V^\lambda \subseteq E_6^-|_{V^\lambda}$, by S_6 -branching to the $V^{(2,2,1)}$ summand where the entry pair $\{6, 7\}$ is forced into column 1.
- (K5) Lemma C: $(R'_4)^2 \cdot V^\lambda \subseteq E_5^-|_{V^\lambda}$, by the analogous S_5 -branching to the $V^{(2,2)}$ -summand where the entry pair $\{5, 6\}$ is forced into column 1.

Positions 12 and 13 vanish via Lemma A; positions 16 and 17 vanish via Lemmas B and C respectively. Computational verification confirms operator vanishing for all four positions at exact rationals (representative value $q = 5/7$), and the empirical fact $S_1 \Omega_{\text{tail}}^{(\lambda)}[\widehat{p}] = 0$ on V^λ holds for all 11 tail positions, suggesting a uniform Strong Pillar 1 statement that remains to be proved at the five open positions.

1 Setup and statement

We use the conventions of [1, 2] throughout. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid/commutation relations. Set $S_i := T_i + 1$; then $S_i^2 = (q+1)S_i$, so S_i has eigenvalues $0, q+1$, and we write $E_i^+ = \ker(T_i - q) = \text{Im}(S_i)$, $E_i^- = \ker(S_i)$.

For $k \geq 1$, $R'_k := S_{k-1} S_{k-2} \cdots S_1$ (with $R'_1 := I$). The staircase chain for S_n is $\Omega := R'_2 R'_3 \cdots R'_n$, the reduced word for the longest element of S_n , of total length $N = \binom{n}{2}$.

For $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ in the Pillar 1 regime ($\ell(\widehat{\lambda}) \geq 2$, every $\widehat{\lambda}_r \geq 2$, $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$), set $\ell := \ell(\widehat{\lambda}) + q$ and $\tau := q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$, and split $\Omega = R \cdot \Omega_{\text{tail}}^{(\lambda)}$ with

$$R = R'_2 \cdots R'_\ell, \quad \Omega_{\text{tail}}^{(\lambda)} = R'_{\ell+1} \cdots R'_n.$$

The Pillar 1 meta-theorem [3] states $\Omega_{\text{tail}}^{(\lambda)} \cdot V^\lambda \subseteq \bigcap_{j=1}^\tau E_j^-|_{V^\lambda}$.

Per-subset operator and Strong Pillar 1. For each tail position $p \in \{\binom{\ell}{2}, \dots, N-1\}$ corresponding to a staircase letter i_p , define the *factor-removed tail chain*

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{p\}}]} := \prod_{k \in \text{tail}, k \neq p} S_{i_k},$$

preserving the staircase order. The *Strong Pillar 1 property at level 1* [2] is the statement

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{p\}}]} \cdot V^\lambda \subseteq E_1^-|_{V^\lambda} \quad \forall p \in \text{tail}. \quad (1)$$

This is the key uniform statement we ultimately want. As shown in [2, Thm. ??], (1) implies that the per-subset operator $M_T(\{p\})$ satisfies $M_T(\{p\})V^\lambda \subseteq E_1^-$, which combined with prefix kill gives $M(S) = 0$ for $|S| = 1$.

Target. We focus on $\lambda = (2, 2, 1, 1, 1)$, so $\widehat{\lambda} = (2, 2)$, $\ell(\widehat{\lambda}) = 2$, $q = 3$, $\ell = 5$, $\tau = 2$, $n = 7$, $N = \binom{7}{2} = 21$. The tail is positions 10–20 with staircase letters $(5, 4, 3, 2, 1; 6, 5, 4, 3, 2, 1)$.

Theorem 1 (Operator vanishing at four positions). *For $\lambda = (2, 2, 1, 1, 1)$, the factor-removed tail chains $\Omega_{\text{tail}}^{(\lambda)[\widehat{\{p\}}]}$ act as the zero operator on V^λ for $p \in \{12, 13, 16, 17\}$.*

This is strictly stronger than Strong Pillar 1 (1) at these four positions (which would only require containment in E_1^-). The proof proceeds via three structural lemmas (A, B, C), each reducing to Pillar 1 on a sub-shape via either embedding or branching.

2 Key identities

Lemma 2 (K1: row-1-pair absorption). *On V^λ for $\lambda = (2, 2, 1, 1, 1)$,*

$$S_1 S_2 = q \cdot I \quad \text{on } E_1^+|_{V^\lambda}.$$

Equivalently, $(S_1 S_2 - qI) \cdot S_1 = 0$ as operators on V^λ .

Proof. $E_1^+|_{V^\lambda} = \ker(T_1 - q)|_{V^\lambda} = \text{Im}(S_1|_{V^\lambda})$, of dimension 4, spanned by Hoefsmit basis vectors e_T for SYTs T with 1, 2 in the same row. Since $\lambda_1 = 2$ and the cell $(2, 2)$ is below $(1, 2)$, this forces 1 at $(1, 1)$ and 2 at $(1, 2)$, with entry 3 at the only legal position $(2, 1)$.

For such T , the axial distance of T at $i = 2$ is $d_T(2) = c_T(3) - c_T(2) = (1 - 2) - (2 - 1) = -2$, never ± 1 . The Hoefsmit 2×2 block at position 2 thus reads

$$T_2 e_T = A_{-2} e_T + e_{s_2 T}, \quad A_{-2} = \frac{q-1}{1-q^{-(-2)}} = \frac{q-1}{1-q^2} = -\frac{1}{q+1},$$

and the swapped tableau $s_2 T$ has 1 at $(1, 1)$, 2 at $(2, 1)$, so $s_2 T \in E_1^-|_{V^\lambda}$ (axial dist $d_{s_2 T}(1) = -1$, same-column). Hence

$$S_2 e_T = \frac{q}{q+1} e_T + e_{s_2 T},$$

with the first summand in E_1^+ and the second in E_1^- . Apply S_1 : $S_1 e_T = (q+1) e_T$ since $e_T \in E_1^+$, and $S_1 e_{s_2 T} = 0$ since $s_2 T \in E_1^-$. Therefore

$$S_1 S_2 e_T = \frac{q}{q+1} (q+1) e_T + 0 = q e_T.$$

By linearity, $S_1 S_2 = qI$ on $E_1^+|_{V^\lambda}$. Since $E_1^+|_{V^\lambda} = \text{Im}(S_1|_{V^\lambda})$, multiplying on the right by S_1 yields the operator identity $(S_1 S_2 - qI)S_1 = 0$. $\square$ $\square$

Lemma 3 (K2: $S_1 R'_n$ collapse). *For any $n \geq 4$, $S_1 R'_n = q \cdot S_{n-1} S_{n-2} \cdots S_3 \cdot S_1$ as operators on V^λ ($\lambda = (2, 2, 1, 1, 1)$).*

Proof. $R'_n = S_{n-1} S_{n-2} \cdots S_2 S_1$. Since S_1 commutes with S_j for $j \geq 3$ ($|1-j| \geq 2$), we can pull S_1 rightward past $S_{n-1}, \dots, S_3$:

$$S_1 R'_n = S_{n-1} S_{n-2} \cdots S_3 \cdot S_1 S_2 S_1.$$

For any $v \in V^\lambda$, $S_1 v \in \text{Im}(S_1|_{V^\lambda}) = E_1^+|_{V^\lambda}$. By Lemma 2 applied to $w = S_1 v$, $S_1 S_2 w = q w$, i.e.

$$S_1 S_2 S_1 v = S_1 S_2 (S_1 v) = q(S_1 v) = q S_1 v.$$

Therefore $S_1 S_2 S_1 = q S_1$ as operators on V^λ , and

$$S_1 R'_n = S_{n-1} \cdots S_3 \cdot S_1 S_2 S_1 = q \cdot S_{n-1} \cdots S_3 \cdot S_1. \quad \square$$

$\square$

3 Structural lemmas

Lemma 4 (Lemma A). $S_1 R'_7 \cdot V^\lambda \subseteq E_3^- \cap E_4^-|_{V^\lambda}$.

Proof. By Lemma 3 (with $n = 7$),

$$S_1 R'_7 \cdot v = q \cdot S_6 S_5 S_4 S_3 \cdot S_1 v$$

for every $v \in V^\lambda$. So

$$S_1 R'_7 \cdot V^\lambda = q \cdot S_6 S_5 S_4 S_3 \cdot E_1^+|_{V^\lambda}.$$

The subspace $E_1^+|_{V^\lambda}$ is spanned by Hoefsmit vectors e_T with 1, 2 at $(1, 1), (1, 2)$; the remaining entries 3, $\dots$, 7 then fill the sub-shape $\nu := (2, 1, 1, 1)$ (cells $(2, 1), (2, 2), (3, 1), (4, 1), (5, 1)$ of λ), according to a standard tableau $\tilde{T}$ of ν with entries $\{3, \dots, 7\}$. Equivalently, with the relabeling $i \leftrightarrow i - 2$ for $i \geq 3$ in T , the map $T \mapsto \tilde{T}$ gives a bijection between SYT of λ in $E_1^+|_{V^\lambda}$ and SYT of $\nu = (2, 1, 1, 1)$.

For $i \geq 3$, the Hecke generator T_i on e_T depends only on the positions of i and $i + 1$ in T , both of which lie in ν . Therefore the action of T_3, T_4, T_5, T_6 on $E_1^+|_{V^\lambda}$ is intertwined by this relabeling bijection with the action of $\tilde{T}_1, \tilde{T}_2, \tilde{T}_3, \tilde{T}_4$ on V^ν in the standard Hoefsmit basis.

Under this identification, the operator $S_6 S_5 S_4 S_3$ on $E_1^+|_{V^\lambda}$ becomes $\tilde{S}_4 \tilde{S}_3 \tilde{S}_2 \tilde{S}_1 = \widetilde{R'_5}$ on V^ν . The shape $\nu = (2, 1^3)$ is a hook with arm $k_h = 2$ and leg $m_h = 3$. Its Pillar 1 parameters are $\widehat{\lambda}(\nu) = (2)$, $\ell(\widehat{\lambda})(\nu) = 1$, $q'(\nu) = 3$, so $\tau(\nu) = q' + 2 - 1 - 2 = 2$ and $\ell(\nu) = 4$, with $\Omega_{\text{tail}}^{(\lambda)}(\nu) = \widetilde{R'_5}$ ($\ell + 1 = 5 = n$, a single block). Pillar 1 (Theorem ??) applied to ν gives

$$\widetilde{R'_5} \cdot V^\nu \subseteq \widetilde{E_1^-} \cap \widetilde{E_2^-}|_{V^\nu}.$$

By the relabeling, $\widetilde{E_1^-} = E_3^-$ and $\widetilde{E_2^-} = E_4^-$ inside V^λ . Therefore

$$S_6 S_5 S_4 S_3 \cdot E_1^+|_{V^\lambda} \subseteq E_3^- \cap E_4^-|_{V^\lambda},$$

and $S_1 R'_7 \cdot V^\lambda$ is a scalar multiple of this, hence contained in the same intersection. $\square$ $\square$

Lemma 5 (Lemma B). $(R'_5)^2 \cdot V^\lambda \subseteq E_6^-|_{V^\lambda}$.

Proof. $R'_5 \in H_q(S_5)$, so $(R'_5)^2 \in H_q(S_5) \subseteq H_q(S_6)$ preserves the $H_q(S_6)$ -branching $V^\lambda \downarrow S_6 = W_1 \oplus W_2$, where $W_1 = V^{(2,1,1,1,1)}$ (the summand with 7 at $(2, 2)$ in V^λ) and $W_2 = V^{(2,2,1,1)}$ (7 at $(5, 1)$).

On W_1 : $W_1 \downarrow S_5 = V^{(1^5)} \oplus V^{(2,1,1,1)}$ corresponding to removable cells (1, 2) and (5, 1) of (2, 1, 1, 1, 1).

- On the sign-representation summand $V^{(1^5)}$: every S_i acts as 0, so $R'_5 \equiv 0$ there, hence $(R'_5)^2 V^{(1^5)} = 0$.
- On $V^{(2,1,1,1)}$: this is a hook (arm 2, leg 3) viewed as an $H_q(S_5)$ -module. By Pillar 1 with $\widehat{\lambda} = (2)$, $q' = 3$, $\tau = 2$, the chain R'_5 (= the Pillar 1 tail $\Omega_{\text{tail}}^{(\lambda)(2,1,1,1)}$ in S_5) sends $V^{(2,1,1,1)}$ into $E_1^-|_{V^{(2,1,1,1)}}$. Then R'_5 applied again ends in S_1 , which annihilates E_1^- . Hence $(R'_5)^2 V^{(2,1,1,1)} = 0$.

Combining, $(R'_5)^2 W_1 = 0$.

On W_2 : $W_2 \downarrow S_5 = V^{(2,1,1,1)} \oplus V^{(2,2,1)}$ corresponding to removable cells (2, 2) and (4, 1) of (2, 2, 1, 1).

- On $V^{(2,1,1,1)}$: same argument as above, $(R'_5)^2 V^{(2,1,1,1)} = 0$.
- On $V^{(2,2,1)}$: in this $H_q(S_5)$ -summand, the entries 6, 7 are positioned at (4, 1) and (5, 1) respectively (from the iterated branching trace). In particular, 6 and 7 lie in the *same column*, so the Hoefsmit action of T_6 is $T_6 e_T = -e_T$ (axial distance -1), giving $S_6 e_T = (T_6 + I)e_T = 0$ on the entire $V^{(2,2,1)}$ -summand.

Therefore $S_6 \cdot (R'_5)^2 W_2 \subseteq S_6 V^{(2,2,1)} = 0$.

Combining W_1 and W_2 : $S_6 \cdot (R'_5)^2 V^\lambda = 0$, i.e., $(R'_5)^2 V^\lambda \subseteq \ker S_6 = E_6^-$. $\square$ $\square$

Lemma 6 (Lemma C). $(R'_4)^2 \cdot V^\lambda \subseteq E_5^-|_{V^\lambda}$.

Proof. Analogous to Lemma 5. $R'_4 \in H_q(S_4) \subseteq H_q(S_5)$ preserves the $H_q(S_5)$ -branching of V^λ . By the iterated branching trace,

$$V^\lambda \downarrow S_5 = V^{(1^5)} \oplus 2V^{(2,1,1,1)} \oplus V^{(2,2,1)},$$

where the $V^{(2,2,1)}$ -summand has entries 5, 6, 7 at positions (3, 1), (4, 1), (5, 1) respectively (deepest column-1 stack). On the sign and hook summands, the previous argument gives $(R'_4)^2 = 0$:

- $V^{(1^5)}$: trivially.
- $V^{(2,1,1,1)}$: R'_4 is the Pillar 1 tail in $H_q(S_4)$ acting on $V^{(2,1,1,1)} \downarrow S_4$, and using hook column-1 plus S_1 -end kill, $(R'_4)^2 = 0$.
- $V^{(2,2,1)}$: this is an $H_q(S_5)$ -module restricted to $H_q(S_4)$: $V^{(2,2,1)} \downarrow S_4 = V^{(2,1,1)} \oplus V^{(2,2)}$ from removable cells (2, 2), (3, 1). On $V^{(2,1,1)}$ (hook), $(R'_4)^2 = 0$ by the same chain argument. On $V^{(2,2)}$: in this sub-summand the entries 5, 6, 7 are at positions (3, 1), (4, 1), (5, 1) in λ ; in particular 5, 6 are same-column.

On the $V^{(2,2)}$ -summand: $T_5 e_T = -e_T$ (same-column for 5, 6), so $S_5 e_T = 0$ there. Therefore $S_5 \cdot (R'_4)^2 V^\lambda = 0$, equivalently $(R'_4)^2 V^\lambda \subseteq E_5^-|_{V^\lambda}$. $\square$ $\square$

4 Proof of Theorem 1

4.1 Positions 12 and 13 via Lemma A

Lemma 7 (Position 12). $\Omega_{\text{tail}}^{(\lambda)[\widehat{\{12\}}]} = 0$ on V^λ .

Proof. Position 12 is the third letter of $R'_6 = S_5 S_4 S_3 S_2 S_1$, which is S_3 . Removing it gives $R'_6[\widehat{S_3}] = S_5 S_4 S_2 S_1$, and

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{12\}}]} = (S_5 S_4 S_2 S_1) \cdot R'_7.$$

Reading the chain $S_5 S_4 S_2 S_1 R'_7$ from the right (operator application order), the factors apply in turn: R'_7, S_1, S_2, S_4, S_5 .

For any $v \in V^\lambda$, set $w := R'_7 v$ and consider the partial product $S_2 S_1 \cdot w = S_2(S_1 w)$. By Lemma 4, $S_1 R'_7 V^\lambda \subseteq E_4^-|_{V^\lambda}$, so $S_1 w \in E_4^-$. Since $|2 - 4| = 2$, S_2 commutes with T_4 , hence preserves $E_4^- = \ker S_4$. Therefore

$$S_2 S_1 R'_7 v \in E_4^-|_{V^\lambda}.$$

Applying S_4 : $S_4 \cdot S_2 S_1 R'_7 v = 0$. The two remaining left factors S_5 do not change this:

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{12\}}]} v = S_5 \cdot S_4 \cdot S_2 \cdot S_1 \cdot R'_7 v = S_5 \cdot 0 = 0. \quad \square$$

□

Lemma 8 (Position 13). $\Omega_{\text{tail}}^{(\lambda)[\widehat{\{13\}}]} = 0$ on V^λ .

Proof. Position 13 is the fourth letter of R'_6 , which is S_2 . Removing it gives $R'_6[\widehat{S_2}] = S_5 S_4 S_3 S_1$, and

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{13\}}]} = (S_5 S_4 S_3 S_1) \cdot R'_7.$$

For any $v \in V^\lambda$, set $w := R'_7 v$. By Lemma 4, $S_1 w \in E_3^-$. Applying S_3 : $S_3 \cdot S_1 w = 0$. Therefore

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{13\}}]} v = S_5 S_4 \cdot S_3 \cdot S_1 \cdot R'_7 v = S_5 S_4 \cdot 0 = 0. \quad \square$$

□

4.2 Positions 16 and 17 via Lemmas B and C

Lemma 9 (Position 16). $\Omega_{\text{tail}}^{(\lambda)[\widehat{\{16\}}]} = 0$ on V^λ .

Proof. Position 16 is the second letter of $R'_7 = S_6 S_5 S_4 S_3 S_2 S_1$, which is S_5 . Removing it leaves $R'_7[\widehat{S_5}] = S_6 S_4 S_3 S_2 S_1 = S_6 R'_5$, where S_6 commutes with each factor of $R'_5 = S_4 S_3 S_2 S_1$ (all distances ≥ 2). Therefore

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{16\}}]} = R'_6 \cdot S_6 R'_5 = R'_6 \cdot R'_5 \cdot S_6.$$

Using $R'_6 = S_5 R'_5$:

$$R'_6 R'_5 = S_5 \cdot (R'_5)^2.$$

Combining and commuting S_6 left past $(R'_5)^2$ (which contains only S_4, S_3, S_2, S_1):

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{16\}}]} = S_5 \cdot (R'_5)^2 \cdot S_6 = S_5 S_6 \cdot (R'_5)^2.$$

For any $v \in V^\lambda$, by Lemma 5, $(R'_5)^2 v \in E_6^-|_{V^\lambda} = \ker S_6$. Hence $S_6 \cdot (R'_5)^2 v = 0$, and $\Omega_{\text{tail}}^{(\lambda)[\widehat{\{16\}}]} v = S_5 \cdot 0 = 0$. □

Lemma 10 (Position 17). $\Omega_{\text{tail}}^{(\lambda)[\widehat{\{17\}}]} = 0$ on V^λ .

Proof. Position 17 is the third letter of R'_7 , namely S_4 . Removing it leaves $R'_7[\widehat{S_4}] = S_6 S_5 S_3 S_2 S_1 = S_6 S_5 \cdot R'_4$ (where $R'_4 = S_3 S_2 S_1$). Therefore

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{17\}}]} = R'_6 \cdot S_6 S_5 R'_4.$$

Commute R'_4 left through S_5 and S_6 (distances ≥ 2 from $\{3, 2, 1\}$): $R'_4 \cdot S_5 = S_5 \cdot R'_4$, $R'_4 \cdot S_6 = S_6 \cdot R'_4$. So

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{17\}}]} = R'_6 R'_4 \cdot S_6 S_5.$$

Using $R'_6 = S_5 S_4 R'_4$ (since $R'_5 = S_4 R'_4$ and $R'_6 = S_5 R'_5$):

$$R'_6 R'_4 = S_5 S_4 \cdot R'_4 \cdot R'_4 = S_5 S_4 \cdot (R'_4)^2,$$

so

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{17\}}]} = S_5 S_4 \cdot (R'_4)^2 \cdot S_6 S_5.$$

Commute S_6 and the trailing S_5 leftward past $(R'_4)^2$ (all commute): $S_6 \cdot (R'_4)^2 = (R'_4)^2 \cdot S_6$, $S_5 \cdot (R'_4)^2 = (R'_4)^2 \cdot S_5$. Inserting these in turn:

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{17\}}]} = S_5 S_4 \cdot S_6 \cdot (R'_4)^2 \cdot S_5 = S_5 S_4 S_6 S_5 \cdot (R'_4)^2.$$

For any $v \in V^\lambda$, the rightmost factor applied is $(R'_4)^2$, and by Lemma 6, $(R'_4)^2 v \in E_5^-|_{V^\lambda} = \ker S_5$. The next applied factor is S_5 , which annihilates this:

$$\Omega_{\text{tail}}^{(\lambda)[\widehat{\{17\}}]} v = S_5 S_4 S_6 \cdot S_5 \cdot (R'_4)^2 v = S_5 S_4 S_6 \cdot 0 = 0. \quad \square \quad \square$$

5 Computational verification and status of open positions

The following table summarizes the 11 tail positions of $\lambda = (2, 2, 1, 1, 1)$ ($\tau = 2$, tail length 11). Ranks and containments are computed exactly at $q = 5/7$ in the Hoefsmit basis. (Equivalent results hold at $q = 2/3$.)

Table 1: Strong Pillar 1 status across all 11 tail positions at $\lambda = (2, 2, 1, 1, 1)$. “Rank” is $\text{rk } \Omega_{\text{tail}}^{(\lambda)[\widehat{\{p\}}]}|_{V^\lambda}$. Empirically, $S_1 \cdot \Omega_{\text{tail}}^{(\lambda)[\widehat{\{p\}}]} = 0$ on V^λ for all 11 positions, which is the Strong Pillar 1 statement at level 1.

p	letter	rank	$\subseteq E_1^-$	operator vanishing	structural proof
10	S_5	1	yes	no	[2, Lem. ??]
11	S_4	1	yes	no	open
12	S_3	0	yes	yes	Lem. 7 (this paper)
13	S_2	0	yes	yes	Lem. 8 (this paper)
14	S_1	1	yes	no	open
15	S_6	1	yes	no	[2, Lem. ??]
16	S_5	0	yes	yes	Lem. 9 (this paper)
17	S_4	0	yes	yes	Lem. 10 (this paper)
18	S_3	1	yes	no	open
19	S_2	1	yes	no	open
20	S_1	1	yes	no	open

Open positions. For positions 11, 14, 18, 19, 20 (rank 1), the operator $\Omega_{\text{tail}}^{(\lambda)[\widehat{p}]}$ does *not* vanish on V^λ (its image is a rank-1 subspace contained in $E_1^-|_{V^\lambda}$), but the structural proof of Strong Pillar 1 (1) at these positions remains open. Computationally:

- All five rank-1 images $\Omega_{\text{tail}}^{(\lambda)[\widehat{p}]}V^\lambda$ are supported on single SYTs (in the Hoefsmit basis), specifically: $p = 10 \rightarrow T_9$, $p = 11 \rightarrow T_5$, $p = 14 \rightarrow T_8$, $p = 15 \rightarrow T_5$, $p = 18 \rightarrow T_8$, $p = 19 \rightarrow T_8$, $p = 20 \rightarrow T_8$. In each case, the supporting SYT has 6 and 7 same-column (column 1), but 1 and 2 at $(1, 1), (1, 3)$ or similar — not the row-1 pattern that triggers our Lemma A bypass.
- The same key identity (Lemma 3) immediately yields $S_1 \cdot \Omega_{\text{tail}}^{(\lambda)[\widehat{p}]} = 0$ *empirically* for all 11 positions, but the structural proof is direct only when the chain can be factorized to expose an inner S_1 -collapse identity (as in Lemmas A–C).

Sketch of a uniform conjectural statement. Define $\mathcal{N}_p := \Omega_{\text{tail}}^{(\lambda)[\widehat{p}]}V^\lambda$. The empirical data suggests

$$S_1 \cdot \mathcal{N}_p = 0 \quad \forall p \in \text{tail}, \quad (2)$$

i.e., Strong Pillar 1 at level 1. By the linear-combination identity [2, Prop. ??], $M_T(\{p\}) = \frac{1}{q+1}\Omega_{\text{tail}}^{(\lambda)[\widehat{p}]} - \frac{1}{(q+1)|\text{tail}|}\Omega_{\text{tail}}^{(\lambda)}$ modulo a global scaling, so once (2) is proved, per-subset Pillar 1 follows for $|S_T| = 1$, closing 4944– (boundary outliers) of the $M(S) = 0$ cases for $|S| \leq 1$ at $m = 3$.

6 Status

Established (this paper).

- Lemma 2: $S_1 S_2 = qI$ on $E_1^+|_{V^\lambda}$ for $\lambda = (2, 2, 1, 1, 1)$, via direct Hoefsmit calculation using axial distance -2 .
- Lemma 3: $S_1 R'_n = q S_{n-1} \cdots S_3 \cdot S_1$ on V^λ for $n \geq 4$.
- Lemma 4: $S_1 R'_7 V^\lambda \subseteq E_3^- \cap E_4^-$, via embedding of $E_1^+|_{V^\lambda}$ into the sub-shape $V^{(2,1,1,1)}$ and Pillar 1 for the hook $(2, 1^3)$.
- Lemma 5: $(R'_5)^2 V^\lambda \subseteq E_6^-$, via S_6 -branching and same-column kill on $V^{(2,2,1)}$.
- Lemma 6: $(R'_4)^2 V^\lambda \subseteq E_5^-$, via S_5 -branching to $V^{(2,2)}$.
- Theorem 1: $\Omega_{\text{tail}}^{(\lambda)[\widehat{p}]} = 0$ on V^λ for $p \in \{12, 13, 16, 17\}$.

Open gaps (precisely stated).

- Strong Pillar 1 at the five rank-1 positions $p \in \{11, 14, 18, 19, 20\}$: the image is verified to lie in E_1^- at exact rationals, and the empirical $S_1 \cdot \Omega_{\text{tail}}^{(\lambda)[\widehat{p}]} = 0$ identity holds, but the structural proof requires a different mechanism (the leftmost-factor / branching reduction used for $p \in \{12, 13, 16, 17\}$ does not directly apply).
- Per-subset Pillar 1 outliers for $|S_T| = 2$: the three subsets $\{11, 12\}, \{12, 15\}, \{15, 16\}$ identified in [1, 2], where $\tau - |S_T| = 0$ and refined Pillar 1 makes no claim. The inductive-kernel mechanism remains case-by-case.
- Extension to $\lambda = (2, 2, 1^m)$ for $m \geq 4$: computational sweep beyond $m = 3$ and inductive argument across m both open.

Methodological note. The key insight enabling the proofs of positions 12, 13 is the identity $S_1 S_2 = qI$ on $E_1^+|_{V^\lambda}$, which arises directly from the Hoefsmit axial distance $d = -2$ for the unique non-degenerate 2×2 block at position 2 inside $E_1^+|_{V^\lambda}$. This identity propagates through the R'_n -chain via Coxeter commutation, ultimately reducing $S_1 R'_n V^\lambda$ to a Pillar 1 computation on the sub-shape $V^{(2, 1^{n-4})}$. This is the first instance, in our program, of a per-subset chain reduction by explicit Hoefsmit identity rather than by chain factorization or full branching; it is likely to extend to other off-hook shapes whose first column hosts a “row-1-pair-plus-sub-hook” decomposition.

The remaining five open positions all involve dropping a letter from R'_6 or R'_7 at a position where the resulting chain factor structure does not admit a clean $(R'_k)^2$ or Lemma A reduction. A uniform mechanism — perhaps via the linear-combination identity of [2, Prop. ??] applied in tandem with the chain operator algebra — is the natural next target.

References

- [1] Clio, *Strong operator vanishing of the staircase chain on $V^{(2, 2, 1^m)}$ (the $|S| = 0$ case of strong per-subset vanishing)*, 28 May 2026.
~/projects/proofs/2026-05-28-strong-per-subset-operator-vanishing.tex.
- [2] Clio, *A linear-combination identity for per-subset Pillar 1: reducing strong operator vanishing to a single tail statement*, 28 May 2026.
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- [3] Clio, *Extended sign-kill for general multi-row $\hat{\lambda}$: the meta-theorem at arbitrary $\ell(\hat{\lambda}) \geq 2$* , 15 May 2026.
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- [4] Clio, *Sharpness of the multi-row containment at $\tau + 1$: $B^{(\lambda)} \cap E_{\tau+1}^-|_{V^\lambda} = 0$* , 17 May 2026.
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